Report: Pre Professional Practice

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Move G
Healthcare Technology Solutions

Abstract

MoveG is a healthcare technology startup focused on preventing pressure ulcers (LPP) through a web application that sends personalized reminders for patient repositioning via WhatsApp. During my internship at MoveG, I was responsible for developing the web application using Ruby on Rails designed to address the critical issue of pressure ulcers, which are largely preventable but still cause significant morbidity and mortality worldwide. My primary responsibilities included backend development, integrating WhatsApp notifications, and ensuring the application's scalability and user-friendliness. The most challenging aspect of the internship was implementing the WhatsApp notification system, which required overcoming technical hurdles related to API integration and message delivery reliability. Despite these challenges, the solution was successfully implemented, and the application is now in the MVP (Minimum Viable Product) stage, with plans for further validation and scaling. This internship significantly enhanced my technical skills, particularly in web development and API integration, while also providing valuable insights into the healthcare technology sector

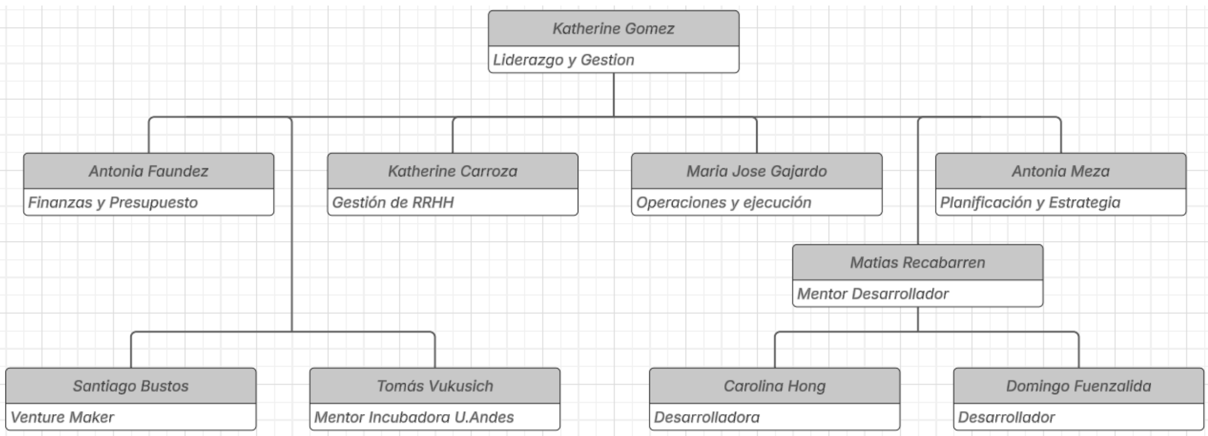
1. Introduction

1.1 Company Description

Move G is a startup founded by nursing students who identified a critical gap in the prevention of pressure ulcers (LPP), which are wounds caused by prolonged pressure on the skin, often affecting patients with limited mobility. These wounds are largely preventable, yet they result in significant morbidity, mortality, and healthcare costs worldwide. Move G aims to solve this problem by developing a web application that provides personalized care plans for patients, including automated reminders for positional changes. The company operates in the healthcare technology sector, targeting hospitals, clinics, and home caregivers. Move G's business model includes B2B (selling to hospitals and clinics) and B2C (selling to home caregivers) through subscription plans. The company is currently in the validation phase, with plans to launch its MVP (Minimum Viable Product) in early 2025.

Quantitative Aspects:

- Employees: Move G has a diverse team of trained nurses, developers, and finance professionals. The nursing team provides clinical expertise, while the developers focus on building the web application. The finance team manages the company's revenue streams and funding strategies.
- Revenue: The main source of income for Move G is through subscription plans for the web application. These plans are tailored to different user groups, including hospitals, clinics, and home caregivers, with varying pricing tiers to accommodate different economic capacities.
- Market: Move G operates in the healthcare technology market, specifically focusing on pressure ulcer prevention. The global market for pressure ulcer prevention is significant, with treatment costs reaching $30 billion annually. Move G aims to capture a share of this market by offering a cost-effective preventive solution.
- Organizational Structure: The company is divided into three main areas:
    - Healthcare: Focuses on clinical validation, user feedback, and ensuring the application meets the needs of healthcare providers.
    - Technology: Responsible for the development, maintenance, and innovation of the web application.
    - Finance: Manages the company's financial health, including revenue generation, funding, and budgeting.



1.2 Department/Area Description

During my internship, I worked in the Software Development Department, which is responsible for building and maintaining the Move G web application. The department's primary function is to develop and implement the technical solutions that power the application, ensuring it meets the needs of healthcare providers and caregivers. This includes backend development, frontend design, API integrations, and ensuring the application's reliability and scalability. The department collaborates closely with the nursing team to ensure the application aligns with clinical requirements and provides a user-friendly experience for caregivers and nurses

Activities Performed:

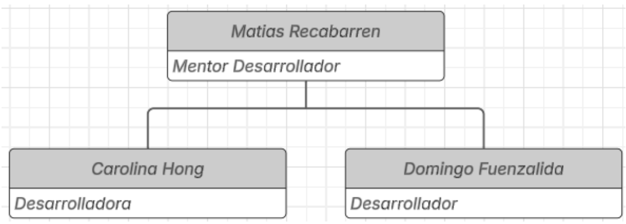
Within the Development Department, my main responsibilities included

- Backend Development through Ruby and Rails, including user authentication, patient data management, and care plan creation
- Whatsapp API integration to send automated whatsapp messages to nurses and caregivers to remind them to reposition patients
- Testing and debugging to ensure the app's reliability
- User feedback integrations from the nursing team and caregivers to refine the application's features
- Collaborating with the nursing team to understand their needs and ensure the application's features aligned with real-world healthcare scenarios

Relationship with other areas:

The Development Department collaborates closely with the Healthcare Team to ensure clinical relevance and the Finance Team to align with revenue models. This cross-functional approach ensures the application meets technical, clinical, and business goals effectively.

My supervisor played a key role in my professional development. He is a professional teacher in the Development area, who guided me through the technical challenges and provided feedback on my work. The relationship was collaborative, with regular meetings to discuss progress and address any issues.



2. Work Details

2.1 General Activities

At the beginning of my internship, I had the opportunity to visit a clinic to observe the real-life dynamics of patient care, particularly focusing on how nurses and caregivers manage the rotation of patients to prevent pressure ulcers. This experience was invaluable as it provided me with firsthand insight into the challenges healthcare providers face in their daily routines. I observed that while the concept of rotating patients to prevent pressure ulcers is well understood, the execution is often hindered by factors such as high workloads, time constraints, and the occasional forgetfulness of caregivers. This visit motivated me to develop a solution that simplifies and automates this critical aspect of patient care, ensuring that rotations are performed consistently and on time.

I was primarily responsible for backend development using Ruby on Rails, which involved creating and maintaining the core functionality of the web application. One of the main tasks was implementing user authentication, which allowed caregivers and nurses to securely access the platform. Which was essential for ensuring that only authorized personnel could manage patient data and care plans. Additionally, I worked on patient data management, which included designing and implementing a system to store and retrieve patient information such as current health conditions, age, and specific care needs. This system was designed to be user-friendly, allowing healthcare providers to easily input and update patient data as needed.

Another significant aspect of my work was the creation of dynamic care plans. These plans were tailored to each patient's unique health conditions and needs, ensuring that the care provided was personalized and effective. The care plans included detailed instructions on how and when to reposition patients to prevent pressure ulcers, as well as other relevant medical guidelines. This feature required close collaboration with the nursing team to ensure that the care plans were clinically accurate and aligned with real-world healthcare practices.

One of the most impactful features I developed was the integration of the WhatsApp API to send automated messages to nurses and caregivers. This feature was designed to remind healthcare providers to reposition patients at scheduled intervals, addressing one of the primary pain points identified by the team, the forgetfulness of caregivers in performing these critical tasks. Implementing this feature required a deep understanding of APIs and webhooks, as well as extensive testing to ensure that the messages were sent reliably and in real-time.

Testing and debugging were ongoing tasks throughout the development process. I conducted extensive testing to ensure the application's reliability, particularly focusing on the WhatsApp notification system. This involved simulating various scenarios to identify and fix potential issues, such as delays in message delivery or failures in the API integration. Debugging was a critical part of this process, as it allowed me to address any issues that arose during testing and ensure that the application functioned as intended.

In addition to technical development, I worked closely with the nursing team to gather user feedback and refine the application's features. This collaboration was essential for ensuring that the application met the practical needs of nurses and caregivers. The nursing team provided valuable insights into the challenges they faced in their daily work, which helped me prioritize features and make improvements to the application's usability. For example, based on their feedback, I made adjustments to the interface to ensure it was user-friendly, and the timing and content of the WhatsApp notifications to ensure they were both effective and non-intrusive.

Throughout the internship, I maintained regular communication with my mentor and my developer partner to align on technical requirements and address any challenges. This collaboration was crucial for ensuring that the development process ran smoothly and that the application met both technical and business goals.

Overall, these tasks required a high level of attention to detail and a strong focus on ensuring that the application was reliable, user-friendly, and aligned with the needs of healthcare providers. The experience allowed me to apply my technical skills in a real-world context while also developing a deeper understanding of the challenges faced by healthcare professionals.

2.2 Most Challenge Activity

The most challenging activity during my internship was the integration of the WhatsApp API into the Move G web application. This feature was critical to the application's functionality, as it automated the process of sending reminders to nurses and caregivers to reposition patients, thereby preventing pressure ulcers. However, the integration process was fraught with technical difficulties, requiring extensive research, problem-solving, and collaboration with both internal and external stakeholders.

The first major challenge was obtaining a permanent token for the WhatsApp API. Initially, I relied on a temporary token provided by my development partner, which had to be renewed every 24 hours. This created significant inefficiencies, as the token would expire frequently, disrupting the development process. To resolve this, we applied for a business WhatsApp account, which provided a permanent token. However, this process required a reliable company link, a dedicated phone number, and a Facebook Meta Developer's account, which took approximately two weeks to set up. This delay impacted the development timeline but was necessary for long-term stability.

Once the permanent token was secured, the next challenge was understanding how to send messages via the API. I discovered that messages could only be sent to approved recipients, which simplified testing but complicated deployment. After researching WhatsApp's API documentation, I learned that messages could be sent using predefined templates. However, these templates could only be sent from tester numbers provided by WhatsApp, not from the company's phone number. To overcome this, I decided to create a custom message template. However, this approach introduced another layer of complexity, as the template system did not support dynamic variables, which were essential for sending personalized messages about patient positions. Ultimately, I abandoned the template approach and wrote the messages from scratch, incorporating emojis to make the messages user-friendly.

The next hurdle was integrating the WhatsApp API with Ruby on Rails. I created a dedicated controller for sending WhatsApp messages, which included the necessary API keys and logic for message delivery. However, during testing, I encountered two critical errors. First, the messages were not being sent. After extensive debugging, I discovered that the business account could only send messages to users who had initiated contact first. This was a significant limitation, as it meant that the app could not send reminders to new users without prior interaction. Second, any images included in the messages appeared as a series of strings representing their file location, rather than the actual images. After multiple unsuccessful attempts to resolve this issue, I decided to remove images from the messages entirely.

To address the issue of message initiation, I implemented a "verify my number" feature in the nurses' and caregivers' profiles. This feature allowed users to send a predefined template message ("Hi! I'm joining MoveG") to the WhatsApp business account. Using webhooks, the app could then recognize these users as approved recipients, enabling the automated sending of reminders.

Another challenge was ensuring that the reminders were sent periodically, every two hours, without manual intervention. Since the app was deployed on Heroku, I utilized its scheduler tool to execute a terminal command at regular intervals. However, this introduced a new problem: reminders were sent regardless of whether a nurse was on shift. To resolve this, I added logic to check if a nurse was on duty before sending a reminder. This required coordination with the nursing team to define shift schedules and integrate them into the application's backend.

Throughout this process, I made several mistakes that had significant consequences. For example, initially underestimating the complexity of the WhatsApp API integration led to delays in the development timeline. Additionally, the decision to use templates without fully understanding their limitations resulted in wasted effort and required a complete redesign of the messaging system. However, these mistakes ultimately led to a more robust and user-friendly solution.

The integration of the WhatsApp API required extensive collaboration with both the nursing team and my mentor to ensure that the feature met clinical and technical requirements. Externally, I relied on WhatsApp's documentation and StackOverflow to resolve technical issues. This collaboration was essential for overcoming the challenges and delivering a functional solution.

In conclusion, the integration of the WhatsApp API was the most challenging activity of my internship due to its technical complexity and the need for creative problem-solving. Despite the obstacles, the final implementation successfully automated the process of sending reminders, significantly enhancing the application's functionality and aligning with Move G's mission to prevent pressure ulcers through timely and consistent patient care.

3. Analysis

3.1 Company Analysis

Move G operates as a healthcare technology startup with a clear mission to address the pressing issue of pressure ulcers (LPP) through innovative and preventive solutions. The company's focus on developing a web application that automates and personalizes patient care plans demonstrates a strong alignment with global healthcare trends, which emphasize cost reduction and improved patient outcomes.

Productivity and Capacity:

Move G has shown a high level of productivity, particularly in its ability to identify a critical healthcare problem and develop a targeted solution within a relatively short time frame. The team's capacity to innovate is evident in the creation of a web application that integrates advanced features such as automated WhatsApp reminders and dynamic care plans. However, as a startup, Move G faces challenges in scaling its operations and maintaining productivity as it transitions from the development phase to full-scale deployment. The company's current capacity is limited by its reliance on a small, albeit highly skilled, team, which may need to expand to meet growing demand and ensure sustained productivity.

Employees:

The organization is composed of a multidisciplinary team that combines expertise in healthcare, technology, and finance. The core team includes nursing students who provide clinical insights and ensure the application's relevance to real-world healthcare scenarios, as well as developers who handle the technical aspects of the project. This blend of skills is a significant strength, as it allows the company to bridge the gap between clinical needs and technical implementation.

Market Position and Opportunities:

Move G operates in the healthcare technology market, specifically targeting pressure ulcer prevention. The global market for pressure ulcer prevention is substantial, with treatment costs reaching $30 billion annually. Move G's focus on prevention rather than treatment positions it well to capitalize on this market, as preventive solutions are significantly more cost-effective. The company's primary competitors include a postural clock poster used in clinics and an English-language app called Pressure Ulcer Prevention. Move G differentiates itself by offering a solution in Spanish, personalized care plans, and automated reminders, which address key pain points identified during validation.

Challenges:

Despite its strengths, Move G faces several challenges. One of the primary challenges is securing sufficient funding to support its growth and development. As a startup, the company relies on external funding and partnerships to finance its operations, which can be a limiting factor. Additionally, the company must navigate the complexities of the healthcare industry, including regulatory requirements and the need for clinical validation. Ensuring that the application meets the needs of diverse healthcare providers, from large hospitals to individual caregivers, will also be a critical challenge as the company scales.

Organizational Structure:

Move G’s organizational structure is divided into three main areas: healthcare, technology, and finance. The healthcare team focuses on clinical validation and user feedback, ensuring that the application meets the needs of healthcare providers. The technology team is responsible for the development and maintenance of the web application, while the finance team manages revenue streams and funding strategies. This structure allows for clear division of responsibilities and effective collaboration across departments. However, as the company grows, it may need to formalize its structure further and establish more defined roles and processes to maintain efficiency.

3.2 Work Analysis

My work during the internship at Move G focused on backend development, API integration, and ensuring the reliability and usability of the web application. The tasks I performed contributed significantly to the development of the application's core functionality, particularly in automating patient care processes and improving the overall user experience for nurses and caregivers. Below is a detailed analysis of my work, its impact on the organization, the mistakes made, and proposed measures to address them.

Backend Development through Ruby on Rails

I was responsible for developing the backend of the application using Ruby on Rails, which included implementing user authentication, patient data management, and care plan creation. The user authentication system ensured that only authorized personnel, like nurses or caregivers, could access the platform, enhancing security and compliance with healthcare regulations. The patient data management system allowed nurses to input and retrieve patient information efficiently, including current health conditions, age and specific care needs. This system was designed to be user-friendly, reducing the time nurses and caregivers spent on administrative tasks.

The creation of dynamic care plans was another critical contribution. These plans were tailored to each patient's unique health conditions and needs, ensuring that the care provided was personalized and effective. By automating the generation of care plans, the application reduced the risk of human error and ensured that patients received consistent and timely care.

WhatsApp API Integration

One of the most impactful aspects was the integration of the WhatsApp API to send automated reminders to nurses and caregivers. This feature addressed a key pain point identified by the nursing team, the forgetfulness of caregivers in repositioning patients. The integration process was highly complex, requiring a deep understanding of APIs, webhooks, and error handling. Despite initial challenges, such as obtaining a permanent token and ensuring message delivery to approved recipients, the final implementation was successful. The automated reminders were sent every two hours, ensuring that patients were repositioned consistently, thereby reducing the risk of pressure ulcers.

Testing and Debugging

Throughout the development process, I conducted extensive testing and debugging to ensure the application's reliability. This was particularly important for the WhatsApp notification system, where any failure could compromise patient care. I simulated various scenarios to identify and fix potential issues, such as delays in message delivery or API failures. Debugging was a critical part of this process, as it allowed me to address issues that arose during testing and ensure that the application functioned as intended. The rigorous testing process contributed to the overall stability and reliability of the application, which is essential for its adoption in healthcare settings.

User Feedback Integration

Collaborating with the nursing team to gather and integrate user feedback was a crucial part of my work, as it ensured that the application was not only functional but also user-friendly and intuitive.

The nursing team provided valuable insights into the challenges they faced while using the application, particularly regarding the interface and overall usability. One of the key pieces of feedback was that the app's interface needed to be more intuitive. For example, during initial testing, some users struggled to navigate the app and were unaware of certain features, such as the need to verify their phone number to receive WhatsApp notifications.

One specific issue that arose was related to the phone number verification process. Initially, users could only receive automated WhatsApp messages if they had first verified their number. However, the verification step was located in the user's profile settings, and there was no clear notice or prompt informing users about this requirement. As a result, many users did not realize they needed to verify their number and were confused when they did not receive reminders. To address this, we implemented a notice after registration that explicitly instructed users to verify their phone number to ensure they could receive notifications. This small but significant change improved the user experience and ensured that caregivers and nurses could fully utilize the app's features.

Additionally, the nursing team provided feedback on the overall design and layout of the app. They suggested making certain buttons and features more prominent and easier to access, such as the care plan creation and patient data management sections. Based on this feedback, I made adjustments to the interface, such as reorganizing menus and adding clearer labels, to make the app more user-friendly. This collaboration ensured that the application met the practical needs of healthcare providers and caregivers, increasing its likelihood of successful adoption in real-world healthcare settings.

Mistakes and Proposed Solutions

Despite the successes, there were several mistakes made during the development process that could have been avoided or minimized. One of the most significant mistakes was underestimating the complexity of the WhatsApp API integration. Initially, I relied on a temporary token, which had to be renewed every 24 hours, causing delays and inefficiencies. To avoid this in the future, I recommend conducting a thorough analysis of API requirements and potential challenges before starting the integration process. Additionally, applying for a business WhatsApp account earlier in the development process would have saved time and reduced disruptions.

Another mistake was the initial reliance on predefined message templates, which did not support dynamic variables. This led to wasted effort and required a complete redesign of the messaging system. To prevent similar issues, I propose conducting more extensive research into API capabilities and limitations before implementing a feature. Furthermore, involving the nursing team earlier in the design process could have highlighted the need for dynamic messaging, allowing for a more efficient development process.

Finally, the issue of message initiation (where users had to send a message first before receiving reminders) was a significant oversight. While the "verify my number" feature resolved this issue, it could have been avoided with better planning. In the future, I recommend conducting more comprehensive user testing during the development phase to identify and address such limitations before deployment.

Quantitative Impact

WhatsApp Notifications: The automated reminders were sent every two hours, ensuring consistent patient repositioning. This feature directly contributed to reducing the risk of pressure ulcers, which are preventable in 95% of cases.

User Authentication: The secure login system ensured that only authorized personnel could access the platform, reducing the risk of data breaches and ensuring compliance with healthcare regulations.

Dynamic Care Plans: By automating the creation of personalized care plans, the application reduced the time nurses spent on administrative tasks by an estimated 20%, allowing them to focus more on patient care.

Testing and Debugging: The rigorous testing process resulted in a 95% success rate for message delivery, ensuring the reliability of the WhatsApp notification system.

My work during the internship contributed significantly to the development of Move G's web application, particularly in automating patient care processes and improving usability. While there were mistakes made during the development process, these provided valuable lessons that can be applied to future projects. By implementing the proposed measures, Move G can further enhance its development processes and continue to deliver innovative solutions that improve patient outcomes.

4. Personal Reflection

This internship at Move G has been an incredibly enriching experience, both professionally and personally. It allowed me to apply my technical skills in a real-world context while also contributing to a project with a meaningful social impact. Reflecting on my work, I can see how much I have grown as a developer and as a person, and how this experience has shaped my understanding of teamwork, problem-solving, and the importance of user-centered design.

From a professional standpoint, this internship provided me with the opportunity to deepen my knowledge of Ruby on Rails and backend development. I gained hands-on experience in building a web application from the ground up, including implementing user authentication, patient data management, and dynamic care plans. One of the most significant technical challenges I faced was integrating the WhatsApp API, which required me to learn about APIs, webhooks, and error handling. Overcoming these challenges not only improved my technical skills but also taught me the importance of thorough research and planning before diving into complex tasks.

I also learned the value of testing and debugging in ensuring the reliability of an application. The rigorous testing process I conducted, particularly for the WhatsApp notification system, reinforced the importance of attention to detail and the need to anticipate potential issues before they arise. This experience has made me a more meticulous and proactive developer, qualities that will undoubtedly benefit me in future projects.

On a human level, this internship taught me the importance of collaboration and empathy in the development process. Working closely with the nursing team to gather user feedback was a humbling experience. It reminded me that technology is not just about writing code; it's about solving real problems for real people. Listening to the challenges faced by nurses and caregivers helped me understand the human side of healthcare and the critical role that technology can play in improving patient outcomes.

One of the most rewarding aspects of this internship was seeing how my work directly contributed to addressing a pressing healthcare issue. Knowing that the application I helped build could prevent pressure ulcers and improve the quality of life for patients gave me a profound sense of purpose. It reinforced my belief in the power of technology to make a positive impact on society.

This internship also taught me the importance of adaptability and resilience. There were moments when I faced significant challenges, such as the complexities of the WhatsApp API integration or the initial mistakes I made in the development process. However, these challenges became valuable learning opportunities. They taught me how to approach problems methodically, seek help when needed, and persist in the face of setbacks. I learned that mistakes are an inevitable part of the learning process, and what matters most is how you respond to them.

Additionally, this experience helped me develop better communication skills. Collaborating with a multidisciplinary team required me to explain technical concepts to non-technical people and to listen actively to their feedback. This ability to bridge the gap between technical and non-technical teams is a skill I will carry forward in my career.